

Oyendrila Dobe

PH.D. · APPLIED SCIENTIST, AUTOMATED REASONING

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[LinkedIn](#) · [Publications](#)

Formal-methods researcher applying automated reasoning to real software — from model checking probabilistic hyperproperties to building neurosymbolic tools that guard against unwanted agentic behaviour.

EXPERIENCE

2024 — now	Applied Scientist · Automated Reasoning Group • Dogwood — a temporal policy language for expressing guardrails on agentic behaviour; contributing to language development and the interpreter implementation. • Kiro spec validation — optimized the pipeline that detects ambiguity and inconsistency in LLM-generated requirements and surfaces them for clarification; contributed across the pipeline, including an LLM refinement phase built on abductive reasoning. • Equalizer & JVerify — deductive verification and equivalence checking for Java: error-reporting UX, library contracts, a benchmarking toolkit, and bitwise integer handling in the Java-to-Boogie translation.	AWS · Boston
Summer 2023	Applied Scientist Intern · Dafny Extended a tool checking cross-language equivalence of programs. Mentor: Rustan Leino.	AWS · Seattle
Summer 2022	Applied Scientist Intern · Privacy Engineering Prototyped detection of data-residency violations in customer cloud setups via automated reasoning. Mentor: Byron Cook.	AWS · New York
2020 — 2024	Research Assistant · Trustworthy & Reliable Technologies Lab • Proposed and proved solutions across the verification of probabilistic hyperproperties — nondeterminism, rewards, asynchronicity, parameter synthesis — and built HyperProb and companion prototypes, with the case studies and experiments that validate them.	Michigan State University
2018 — 2022	Teaching Assistant Computer-Aided Verification (grad), Data Structures, OOP, Discrete Math. Lead TA for a 194-student class.	Michigan State University

EDUCATION

2018 — 2024	Ph.D., Computer Science <i>Verification of Probabilistic Hyperproperties on Markov Models</i> — advisor Prof. Borzoo Bonakdarpour.	Michigan State University
2013 — 2017	B.Tech., Computer Science & Engineering Thesis: detection and extraction of tumors from brain MRI scans.	MAKAUT, Kolkata

SELECTED PUBLICATIONS

1. **Lightweight verification of hyperproperties.** Dobe et al. ATVA 2023.
2. **Introducing asynchronicity to probabilistic hyperproperties.** Gerlach, Dobe, Abraham, Bartocci, Bonakdarpour. QEST 2023.
3. **Model checking hyperproperties for Markov decision processes.** Information & Computation, vol. 289, 2022.
4. **Probabilistic hyperproperties with rewards.** NASA Formal Methods 2022.
5. **HyperProb: a model checker for probabilistic hyperproperties.** FM 2021.

SERVICE & PEER REVIEW

Program Committee	EMSOFT 2025 · SpecOps 2026
Artifact Evaluation	TACAS 2024–26 · CAV 2024–25 · OOPSLA 2026 · iFM 2025 · PLDI 2023 · QEST 2022–23
Sub-reviewer	CAV 2024 · EMSOFT 2024 · VSTTE 2023 · NFM 2022 · QEST 2022 · DISC 2022 · ATVA 2021